

Exercise 3 - Bubble Sort Complexity Analysis

1. Derivation of Formulas

(i) Number of Comparisons:

In Bubble Sort, the outer loop runs n times. The inner loop runs $n - 1$ times in the first pass, $n - 2$ in the second, and so on until 1 in the last pass.

Total comparisons = $(n - 1) + (n - 2) + \dots + 2 + 1$

$$\text{Comparisons} = \sum_{k=1}^{n-1} k = \frac{n(n-1)}{2}$$

This is the same for best, worst, and average cases since Bubble Sort always compares adjacent elements regardless of whether they will end up being swapped.

(ii) Average-case Number of Swaps:

- **Best case** (already sorted): 0 swaps
- **Worst case** (reverse sorted): $\frac{n(n-1)}{2}$ swaps
- **Average case**: On average, about half the comparisons result in a swap

Therefore:

$$\text{Average swaps} \approx \frac{1}{2} \times \frac{n(n-1)}{2} = \frac{n(n-1)}{4}$$

4. Results and Discussion

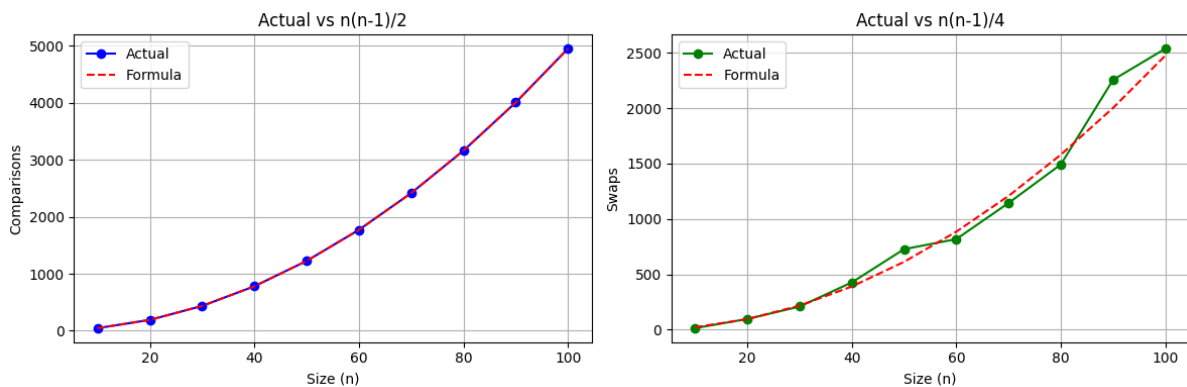


Figure 1: Bubble Sort: Comparisons and Swaps vs Input Size (Average Case)

Discussion:

- **Comparisons:** The actual comparison counts (blue dots) fall exactly on the red curve $\frac{n(n-1)}{2}$. This shows that Bubble Sort always makes $\frac{n(n-1)}{2}$ comparisons regardless of input order.
- **Swaps:** The actual swap counts (green dots) closely follow the red curve $\frac{n(n-1)}{4}$. Small variations are expected due to the random input arrays.
- **Swaps/Comparisons Ratio:** On average, the ratio of swaps to comparisons is about 0.5, confirming that approximately half the comparisons lead to swaps in the average case.
- **Quadratic Growth:** Both plots show clear quadratic growth - as input size doubles, the number of operations roughly quadruples, consistent with $O(n^2)$ complexity.

In conclusion:

$$\text{Comparisons} = \frac{n(n-1)}{2}, \quad \text{Average swaps} \approx \frac{n(n-1)}{4}$$